A computer's hardware devices fall into one of four categories

Essential Computer Hardware

- **Processor:** The procedure that transforms raw data into useful information is called processing. To perform this transformation, the computer uses two components: the processor and memory. The processor is like the brain of the computer; it organizes and carries out instructions that come from either the user or the software.

- **Memory:** In a computer, memory is one or more sets of chips that store data and/or program instructions, cither temporarily or permanently. Memory is a critical processing component in any computer Personal computers use several different types of memory, but the two most important are called random access memory (RAM) and read-only memory (ROM). These two types of memory work in very different ways and perform distinct functions.

- **Input and output devices:** A personal computer would be useless if you could not interact with it because the machine could not receive instructions or deliver the results of its work. Input devices accept data and instructions from the user or from another computer system (such as computer on the Internet). Output devices return processed data to the user or to another computer system.

- **Storage:** A computer can function with only processing, memory, input, and output devices. To be really useful, however; a computer also needs a place to keep program files and related data when they are not in use. The purpose of storage is to hold data permanently, even when the computer is turned off.

Software

Software is a set of instructions that makes the computer perform tasks. In other words, software tells the computer what to do. Some programs exist primarily for the computer's use to help or perform tasks and manage its own resources (**System Software**). Other types of programs exist for the user, enabling him or her to perform tasks such as creating documents. Thousands of different software programs are available for use on personal computers (**Application Software**).

System Software

- **Operating System:** An operating system tells the computer how to use its own components. Examples of operating systems include Windows, the Macintosh Operating System, and Linux. An operating system is essential for any computer; because it acts as an interpreter between the hardware, application programs, and the user.

  When a program wants the hardware to do something, it communicates through the operating system. Similarly, when you want the hardware to do something (such as copying or printing a file), your request is handled by the operating system.

- **Network operating system:** A network operating system allows computers to communicate and share data across a network while controlling network operations and overseeing the network's security.

- **Utility:** A utility is a program that makes the computer system easier to use or performs highly specialized functions. Utilities arc used to manage disks, troubleshoot hardware problems, and perform other tasks that the operating system itself may not be able to do.

5 Operating Systems

By completing this lesson you will learn,

- the necessity of an operating system
- the function of an operating system
- user interfaces of operating systems
- services of an operating system
- types of operating systems
- advantages of operating systems
- utility Programmes of operating system
- drives, folder and files.

5.1 Introduction to Operating Systems

A Computer consists of hardware, firmware and software.

Any physical component of a computer system with a definite shape is called a hardware. Examples of hardware include: mouse, keyboard, display unit, hard disk, speaker, printer etc.

The booting instructions stored in the ROM (Read Only Memory) are called firmware. The initial text information displayed on the screen are displayed by firmware.

How the initial operations of a computer are performed

- When the user powers up the computer the CPU (Central Processing Unit) activates the BIOS (Basic Input Output System).
- The first program activated is POST (Power On Self-Test). Using the CMOS (Complementary Metal Oxide Semiconductor) memory this checks all the hardware and confirms that all are functioning properly.
- After that it reads the MBR (Master Boot Record) in boot drive in accordance with the firmware 'bootstrap loader' which is provided by the computer manufacturer.
- Then the computer loads in the Operating System in boot drive to the RAM (Random Access Memory)

- Once this is performed the Operating System takes over the control of the computer and displays a user interface to the user.

This whole process is called booting which means that an Operating System is loaded into the RAM (main memory).

Software is a set of instructions given to the computer to perform some activity using a computer. There are many types of software. They can be broadly classified as follows:

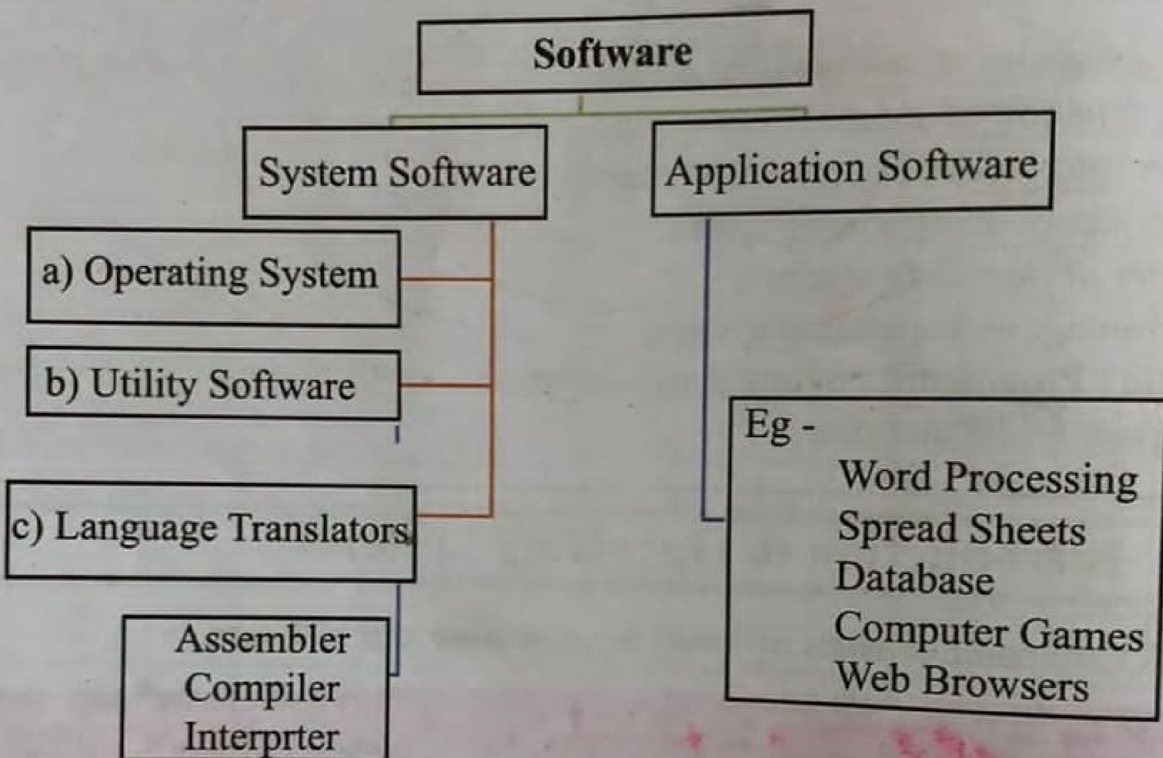

Figure 5.1 - Types of Software

5.1.1 System Software

1) System Software - System software are generally divided into three types. They are:

   a. **Operating System** – The Operating System provides for the user to utilize the functions of a computer by managing the hardware and software in it. The image 5.2 below depicts how the system software and application software interact with the hardware.

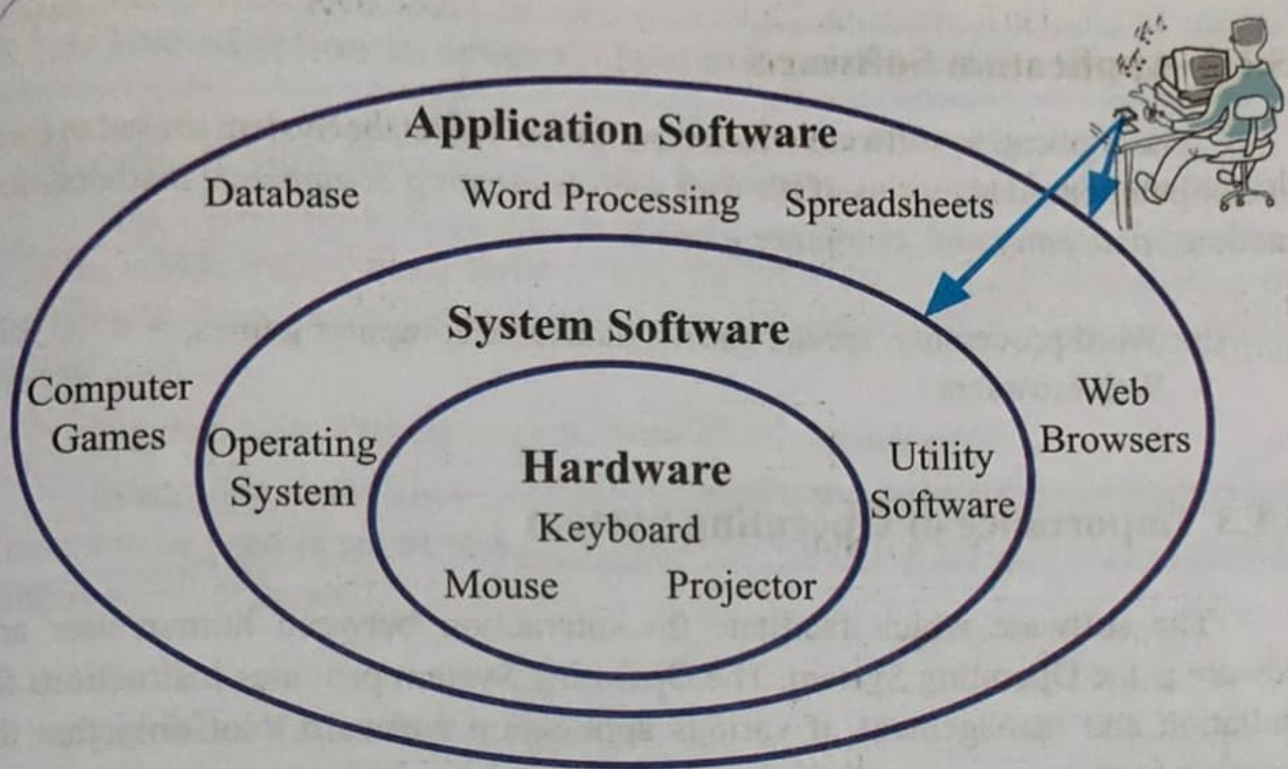

Figure 5.2 – Hardware, System Software, Application Software

b. **Utility Software** – These are used to manage and analyze the software in the computer. The utility software differ from the application software in their complexity and operational activities. Utility software helps in managing the resources of the computer. However, the application software function in different to the utility software. There are many utility software which dedicated to perform certain functions. Some of them are mentioned below:

   i. Anti -Virus Software – to protect the computer from virus infections
   ii. Disk Formatting – to prepare the storage device in order to save the files and folders

c. **Language Translators** - A computer program (software) is made up by using a set of instruction codes. These instructions are written in high level languages which are very close to the human languages. These high level languages are translated into machine language (i.e 0's and 1's) which are understood by the computer by language translators. assembler, compiler and interpreter are examples for language translators.

5.1.2 Application Software

The application software which runs on the Operating System is used to carry out computer based activities of the user such as creating documents, mathematical functions, data entry and, computer games.

Ex: Word processing, spread sheets, database, computer games, Web browsers

5.1.3 Importance of Operating System

The software which facilitate the interaction between human user and hardware is the Operating System. The Operating System provides instructions for installation and management of various application software. Not only that the Operating System manages all the input, output and computer memory too, which means that Operating System is the sole software which manages the whole computer system.

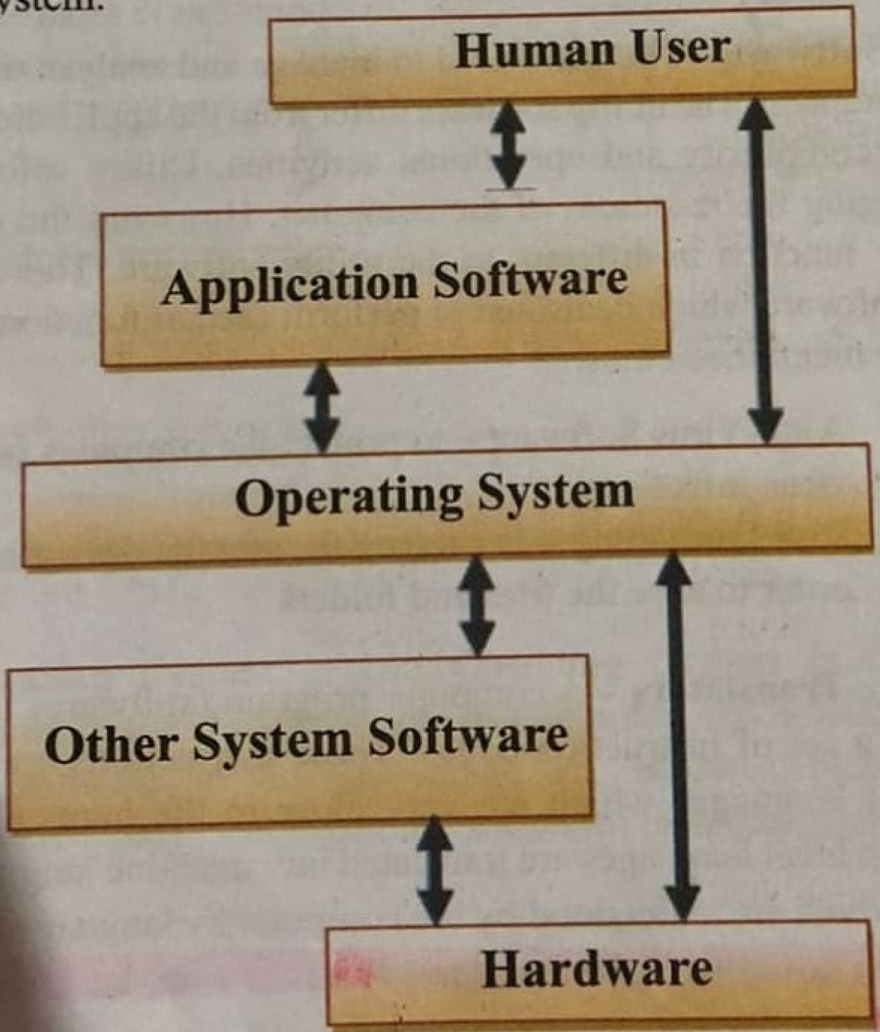

Figure 5.3 - Interaction between the user and the computer

5.1.5 Types of Operating Systems

The main function of an Operating System is to provide an environment suitable for executing the commands issued by the user. Based on the functionality of the Operating System it can be classified as:

1. Single user operating system
2. Multi user operating system
3. Multi-tasking operating system
4. Real time operating system

1. Single user Operating System

The Operating System which provides service to one person at a time is called a Single User Operating System.

Example - MS Dos Operating System

2. Multi-User Operating System

The Operating System which allows multiple users to use a system is simultaneously called a multi-user operating system. This type of Operating System is commonly used in Mainframe or Server computer where several users are connected to a computer system simultaneously.

Example - Linux, Windows server

3. Multi-tasking Operating System

The operating system which allows to run multiple process at the same time is called a multi-tasking operating system. A single user can run multiple operations (tasks) at the same time on this type of operating system.

Example - Windows 7, Windows 8, Ubuntu, Mac Operating System

4. Real Time Operating System

These are the Operating Systems which gives the output in real time without any observable delays. Real Time Operating Systems are mostly utilized in ATM end points. Also these kind of Operating Systems are installed in scientific devices and small gadgets. These Operating Systems are specifically designed for particular devices.

Example - ATM machines, Calculators

5.1.6 Services of an Operating System

The Operating System is a software which manages the hardware and other software in a computer system. It provides services to other software. There are two main services performed by an operating system. They are:

1. Managing the hardware of a computer
2. Providing user interface

1) Managing the Hardware of a computer

Hardware of a computer are managed by using the following processes;

i. Process Management
ii. Memory Management
iii. Device Management
iv. File Management
v. Security Management
vi. Network Management

Figure 5.4 shows the inter connection between resource management within computer.

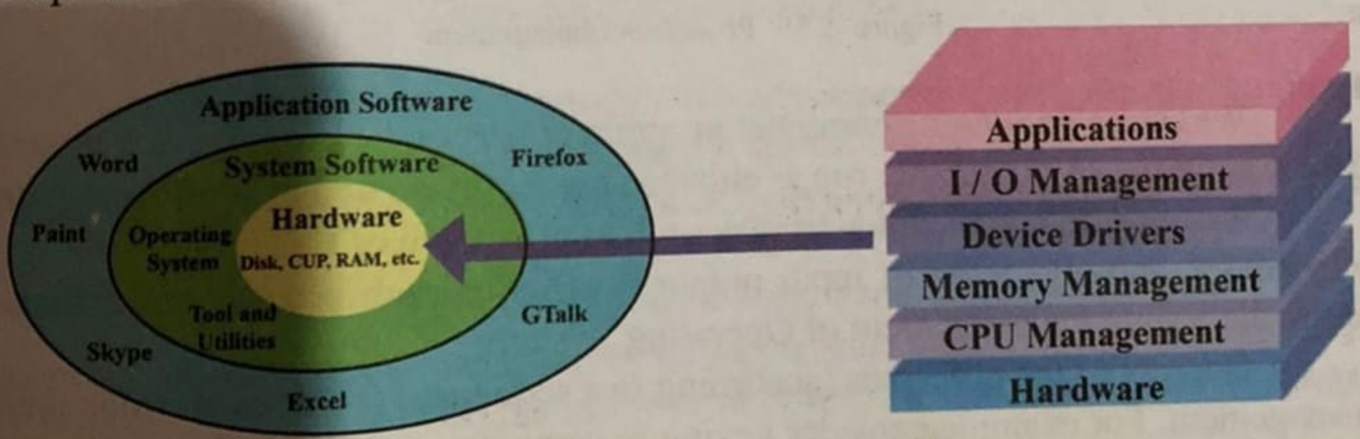

Figure 5.4 - Resource Management

Let us now consider the above mentioned management tasks;

Summary

- Operating System is essential for Operating a Computer.
- All the Application Software in the computer runs on the Operating System.
- The User Interface functions as the facilitator between the computer and the user.
- The GUI is convenient to the user than the CLI.
- The types of Operating System are: Single User, Multi User, Multi-tasking and Real Time User.
- The Operating System which provides service to one person at a time is called a Single User Operating System.
- The Operating System which allows multiple users to use a system is called a Multi User Operating System.
- The Operating System which allows to run multiple process at the same time is called a Multi-Tasking Operating System.
- The Operating System helps in managing all the resources of a computer. The hard disk is partitioned and formatted before installing an Operating System.
- A file consists of a File Name and Extension
- Folders are used to save files
- In order to save the files the user creates folders inside the drive.